

Q1. Create a New Database and Table for Employees

Ans: CREATE DATABASE company_db;

```
Create table employees(  
employee_id int primary key,  
first_name varchar(50),  
last_name varchar(50),  
department varchar(50),  
salary int,  
hire_date date);
```

Q2. Insert Data into Employees Table

```
INSERT INTO employees (employee_id, first_name, last_name, department,  
salary, hire_date)
```

```
VALUES
```

```
(101, 'Amit', 'Sharma', 'HR', 50000, '2020-01-15'),  
(102, 'Riya', 'Kapoor', 'Sales', 75000, '2019-03-22'),  
(103, 'Raj', 'Mehta', 'IT', 90000, '2018-07-11'),  
(104, 'Neha', 'Verma', 'IT', 85000, '2021-09-01'),  
(105, 'Arjun', 'Singh', 'Finance', 60000, '2022-02-10');
```

Q3. Display All Employee Records Sorted by Salary (Lowest to Highest)

Hint: Use the order by clause on the salary column.

Ans: select * from employees order by salary asc;

Q4. Show Employees Sorted by Department (A–Z) and Salary (High → Low)

Ans: select * from employees order by department asc , salary desc;

Q5. List All Employees in the IT Department, Ordered by Hire Date (Newest First)

Ans: select * from employees where department = 'IT' order by hire_date desc;

Q6. Create and Populate a Sales Table Task: Create a table sales to track sales data:

Ans: CREATE TABLE sales (
sale_id INT PRIMARY KEY,
customer_name VARCHAR(100),
amount int,

```
sale_date DATE  
);
```

Insert:

```
INSERT INTO sales (sale_id, customer_name, amount, sale_date)  
VALUES  
(1, 'Aditi', 1500, '2024-08-01'),  
(2, 'Rohan', 2200, '2024-08-03'),  
(3, 'Aditi', 3500, '2024-09-05'),  
(4, 'Meena', 2700, '2024-09-15'),  
(5, 'Rohan', 4500, '2024-09-25');
```

Q7. Display All Sales Records Sorted by Amount (Highest → Lowest)

Ans: select * from sales order by amount desc;

Q8. Show All Sales Made by Customer “Aditi”

Ans: select * from sales where customer_name = “Aditi”;

Q9. What is the Difference Between a Primary Key and a Foreign Key?

Ans: Primary Key:

- A primary key uniquely identifies each record in a table.
- It cannot be NULL.

A table can have only one primary key.

Foreign Key:

- A foreign key links two tables together.
- It refers to the primary key of another table.
- It can have duplicate values.
- It can be NULL depending on constraints.

Q10. What Are Constraints in SQL and Why Are They Used?

What Are Constraints:

Constraints are rules applied to columns or tables in SQL to control the type of data that can be stored. They ensure that the data in the database is accurate, valid, and reliable.

Constraints are used to:

- Prevent invalid data from being inserted
- Maintain data integrity
- Enforce business rules
- Ensure relationships between tables remain correct